3- Project Planning & Management

Project Title

Ai Impact On Jobs Market

1. Project Idea and Objectives

Project Idea

The project aims to analyze the impact of Artificial Intelligence (AI) on the job market by examining employment trends, job automation risks, changes in skill requirements, and the emergence of new job opportunities.

The project will use data analysis and visualization techniques to understand how AI may transform different occupations and industries and how individuals and organizations can prepare for these changes.

Project Objectives

- Understand how AI affects different industries and occupations.

- Identify jobs that are more exposed to AI automation.

- Analyze the relationship between AI adoption and employment trends.

- Examine differences in salaries and employment opportunities across industries and occupations, where data is available.

- Identify the skills required to succeed in an AI-driven job market.

- Develop data-driven insights and recommendations to help individuals and organizations prepare for future labor market changes.

- Apply Excel, Power Query, and Power BI to clean, analyze, and visualize the collected data.

2. Project Scope

The project will focus on analyzing job market data related to AI adoption, employment trends, automation risk, salaries, industries, occupations, and required skills.

The analysis will include:

- Comparing the impact of AI across different industries.
- Identifying occupations with high, medium, and low AI exposure, depending on the available data.
- Examining changes in job opportunities and employment trends.
- Analyzing salary differences across occupations and industries, where data is available.
- Exploring changes in required skills and the emergence of new job opportunities.
- Cleaning and transforming data using Excel and Power Query.
- Performing exploratory data analysis and calculating key metrics.
- Building an interactive Power BI dashboard using data modeling, DAX measures, and appropriate visualizations.
- Presenting the main findings, limitations, and recommendations in a final analytical report and presentation.

The project will focus on the available datasets and their limitations. AI exposure will be treated as an indicator of potential job transformation, not proof that a job will disappear. The analysis will distinguish between potential exposure to AI and actual employment losses.

3. Project Timeline

| Phase | Activities | Duration |

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| Phase 1 | Define project objectives, scope, deliverables, and research questions | 1 day |

| Phase 2 | Collect datasets, identify reliable sources, and understand the data | 2 days |

| Phase 3 | Clean and transform data using Excel and Power Query | 2 days |

| Phase 4 | Analyze trends, perform calculations, and identify key findings | 2 days |

| Phase 5 | Build the Power BI data model, create DAX measures, and develop the interactive dashboard | 3 days |

| Phase 6 | Review findings, validate results, prepare the final report and presentation | 1 day |

Total Estimated Duration: 11 working days.

The timeline is an initial estimate and may be adjusted depending on data availability, data quality, and the complexity of the analysis.

4. Roles and Responsibilities

The project will be completed by six team members, with each member responsible for a specific area of work. Team members will coordinate throughout the project to ensure that the final results are accurate and consistent.

4.1 Basmala – Data Collection and Research

- Identify reliable data sources related to AI exposure, employment trends, salaries, industries, occupations, and required skills.

- Collect relevant datasets from reliable sources.

- Review the structure, definitions, and limitations of the collected data.

- Document data sources and references.

- Coordinate with the data preparation team to ensure that the collected datasets are suitable for analysis.

Deliverables : Raw datasets, source references, and an initial description of the data.

4.2 Omar – Data Cleaning and Preparation

- Inspect the collected datasets using Excel.

- Identify and handle missing values, duplicate records, inconsistent formats, and incorrect data types.

- Standardize column names, occupation categories, industry classifications, and other relevant fields.

- Use Power Query to clean, transform, and combine datasets when necessary.

- Validate data quality and document important data limitations.

- Prepare the final dataset for analysis and Power BI.

Deliverables : Cleaned datasets, Power Query transformation steps, and data quality checks.

4.3 Basant – Data Modeling and DAX in Power BI

- Import the cleaned datasets into Power BI.

- Build an appropriate data model and establish relationships between tables where necessary.

- Create DAX measures to calculate the project's key metrics.

- Develop measures such as occupation counts, average salaries, AI exposure percentages, and employment growth indicators, depending on the available data.

- Validate DAX calculations against the Excel analysis.

- Coordinate with the dashboard developer to ensure that the measures support the required visualizations.

Deliverables : Power BI data model, DAX measures, and validated calculations.

4.4 Menna, Mirvat – Power BI Dashboard and Visualization

- Design and develop the interactive Power BI dashboard.

- Select suitable visualizations, including KPI cards, bar charts, tables, and trend charts.

- Add slicers and filters for industries, occupations, AI exposure categories, and other relevant fields where applicable.

- Organize the dashboard into a clear and professional layout.

- Ensure that the dashboard communicates the project's main findings effectively.

- Test the visuals, filters, and interactions and coordinate with the DAX developer to resolve any issues.

Deliverables : Interactive Power BI dashboard and final visual design.

4.5 Neama – Insights, Recommendations, Final Report and Presentation

- Review the analytical results and dashboard findings.

- Identify the main insights regarding AI exposure, employment trends, salaries, and changing skill requirements.

- Develop evidence-based recommendations for individuals, employers, and policymakers.

- Prepare the final PowerPoint presentation, including the project objectives, scope, methodology, analysis, findings, and recommendations.

- Coordinate the preparation of the final analytical report.

- Ensure that the presentation and report use accurate results and properly documented sources.

Deliverables : Key insights, recommendations, final analytical report, and PowerPoint presentation.

Shared Responsibilities

All team members will contribute to the following activities:

- Review the accuracy and consistency of the final results.

- Validate calculations and cross-check dashboard figures.

- Identify limitations in the available data and avoid unsupported conclusions.

- Test the final dashboard and ensure that its filters and visuals work correctly.

- Prepare for the final presentation and participate in answering questions about the project.

5. Potential Risks and Mitigation Strategies

| Potential Risk | Mitigation Strategy |

| Missing or inconsistent data | Clean, validate, and document data quality issues using Excel and Power Query. |

| Limited data availability | Use reliable alternative sources and clearly define the analysis scope. |

| Outdated information | Check publication dates and prioritize recent, relevant data. |

| Different definitions across datasets | Review source methodologies and standardize definitions where appropriate. |

| Misinterpreting AI's impact | Distinguish between potential AI exposure and actual job losses. |

| Incorrect calculations or DAX measures | Validate calculations in Excel and cross-check them against Power BI results. |

| Dashboard errors or misleading visuals | Test filters, relationships, measures, and visualizations before submission. |

| Delays in project completion | Follow the timeline, assign clear responsibilities, and prioritize essential deliverables. |

| Poor coordination between team members | Establish clear handover points and review progress throughout the project. |

| Unsupported conclusions or recommendations | Base conclusions on the available evidence and clearly explain data limitations. |

6. Key Performance Indicators (KPIs)

The project's success will be measured using the following indicators:

- Data Quality Rate : Percentage of records that pass the defined data quality checks.

- Data Completeness Rate : Percentage of required fields containing valid data.

- Industry Coverage : Number of industries included in the analysis.

-Occupation Coverage : Number of occupations analyzed.

- AI Exposure Distribution : Percentage of occupations in each AI exposure category, where the dataset provides this information.

- Employment Trend Analysis : Changes in employment levels or job growth over the available period.

- Salary Analysis Coverage : Number or percentage of occupations and industries with usable salary data.

- Data Preparation Completion : Completion of the required cleaning and transformation steps.

- DAX Validation : Confirmation that key Power BI measures are consistent with independently checked calculations.

- Dashboard Completion : Delivery of a functional, interactive Power BI dashboard.

- Report and Presentation Completion : Completion of the analytical report and final presentation.

- Project Completion Rate : Percentage of planned deliverables completed within the agreed timeline.

Where applicable, the team will define measurable targets for these indicators after reviewing the available datasets.

7. Expected Outcome

The project is expected to deliver an interactive Power BI dashboard and a final analytical report explaining how AI may transform different occupations, influence employment trends, affect salary patterns, and change the skills required in the future job market.

The final deliverables will include:

- A cleaned and prepared dataset suitable for analysis.

- An Excel analysis file containing preliminary calculations, comparisons, and findings.

- A Power BI dashboard with interactive visualizations, KPI cards, charts, and filters.

- A final analytical report documenting the methodology, key findings, data limitations, and recommendations.

- A PowerPoint presentation summarizing the project and communicating the main results.

The project will aim to provide evidence-based insights that help individuals, employers, and policymakers better understand the potential implications of AI for the labor market and prepare for future changes.

The conclusions will be limited to what the available data supports, and potential AI exposure will not automatically be interpreted as evidence of job elimination.